

*****Default*****
pico-ice compiled pi5-70 executed pi3-28
10/26/25
*****Default*****

```
devel@pi5-70:~/PipelineC $ git log
commit 422abd8ec3496d109e1153989ac848503e1292fe (HEAD -> test-dev, origin/test-dev)
Merge: ba22d9f 0213aad
Author: develone <develone@sbcglobal.net>
Date: Thu Apr 24 15:12:02 2025 -0600
```

Merge branch 'JulianKemmerer:master' into test-dev

```
devel@pi5-70:~/pico-ice/PipelineC/pmod-ethernet/examples/net $ git log
commit 854eef3655f9cf3b912ab83cd14d646fdadd43 (HEAD -> test-dev-100424, origin/test-dev-100424)
Author: Edward Vidal Jr <develone@sbcglobal.net>
Date: Wed Oct 22 09:49:13 2025 -0600
```

Testing Google Cloud Free
Generating test-top.uf2 for the pico-ice
RP2040 & iCE40UP5K FPGA
leddigits
PipelineC blink-ex
PipelineC blink-ex-uart-echo
10/22/25

```
devel@pi5-70:~/pico-ice/PipelineC/pmod-ethernet/examples/net $ ./compile-work_test.sh
```

```
devel@pi5-70:~/pico-ice/PipelineC/pmod-ethernet/examples/net $ cp work_test ~/pico-ice-a0.b1.c2.d3.e4.f5/
```

```
devel@pi5-70:~/pico-ice/PipelineC/pmod-ethernet/examples/pico-ice/ice_makefile_pipelinec $ cp gateway.uf2 ~/pico-ice-a0.b1.c2.d3.e4.f5/
```

```
devel@pi5-70:~/pico-ice/PipelineC/pmod-ethernet/examples/pico-ice/ice_makefile_pipelinec $
rsync -avl --delete /home/devel/pico-ice-a0.b1.c2.d3.e4.f5 pi3-28:~/
sending incremental file list
pico-ice-a0.b1.c2.d3.e4.f5/
pico-ice-a0.b1.c2.d3.e4.f5/gateway.uf2
pico-ice-a0.b1.c2.d3.e4.f5/work_test
```

```
devel@pi3-28:~/pico-ice-a0.b1.c2.d3.e4.f5 $ sudo ./work_test
CPU threads: 1
n 'work()'s: 1
Total tx bytes: 8
Total rx bytes: 4
CPU took 0.001652 seconds to execute
CPU iteration time: 0.001652 seconds
CPU bytes per sec: 7263.912253 B/s
FPGA took 0.001050 seconds to execute
```

FPGA iteration time: 0.001050 seconds

FPGA bytes per sec: 11428.621253 B/s

Speedup: 1.573342

pmod-eth-rx

The image shows the Wireshark network traffic capture interface. The title bar indicates the capture is from the interface 'eth0'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and packet analysis. The filter bar shows the active filter: 'eth.addr == a0:b1:c2:d3:e4:f5'. The packet list pane displays two captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
34	6.623582063	Raspberr_ad:a6:2c	a0:b1:c2:d3:e4:f5	0xffff	22	Ethernet II
36	6.623904138	a0:b1:c2:d3:e4:f5	Raspberr_ad:a6:2c	0xffff	60	Ethernet II

The packet details pane shows the selected packet (No. 34) with the following structure:

- Frame 34: 22 bytes on wire (176 bits) captured (220 bits) on 0
- Ethernet II, Src: Raspberr_ad:a6:2c, Dst: a0:b1:c2:d3:e4:f5
- Data (8 bytes)
- Data: 534d5d56560f235c [Length: 8]

The packet bytes pane displays the raw data in hexadecimal and ASCII:

```

0000  a0 b1 c2 d3 e4 f5 b8 27 eb ad a6 2c ff ff 53 4d  .....',..SM
0010  5d 56 56 0f 23 5c                                ]VW.#\
  
```

pmod-eth-tx

The image shows the Wireshark network protocol analyzer interface. The title bar indicates the capture is from the interface 'eth0'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The packet list pane shows two captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
34	6.623582063	Raspberr_ad:a6:2c	a0:b1:c2:d3:e4:f5	0xffff	22	Ethernet II
36	6.623904138	a0:b1:c2:d3:e4:f5	Raspberr_ad:a6:2c	0xffff	60	Ethernet II

The packet details pane for the selected packet (No. 36) shows the following structure:

- Frame 36: 60 bytes on wire (48 bytes captured)
- Ethernet II, Src: a0:b1:c2 (Raspberry Pi), Dst: a0:b1:c2:d3:e4:f5 (Raspberry Pi)
- Data (46 bytes)

The packet bytes pane displays the raw data in hexadecimal and ASCII. The first 10 bytes are shown in the image:

```

0000  b8 27 eb ad a6 2c a0 b1 c2 d3 e4 f5 ff ff 69 89  .'. . . . .i.
0010  00 5b 00 00 00 00 00 00 00 00 00 00 00 00 00  .[. . . . .
0020  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
  
```